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Characterization of the Intrinsic Neural Timescale Indices as a Reliable Fingerprint of Brain Dynamics

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Abstract

Intrinsic neural timescales (INTs) characterize the persistence of neural activity over time and offer insight into the brain's temporal hierarchy. This thesis evaluates and compares six autocorrelation-based INT metrics—AR(1) Coefficient (Autoregressive Coefficient at 1), First Zero Lag, Lag at ACF = 1/e, Sum of Positive ACFs, ACF at Half Lag, Exponential Decay Constant (τ)—using resting-state fMRI data from 400 cortical regions across a large, lifespan sample (ages 6–85) from the NKI-Rockland dataset. A discriminability analysis revealed that the AR(1) coefficient provided the highest test-retest reliability and individual specificity across repeated scans, establishing it as the most reliable INT estimator. Using the AR(1) metric, we investigated how intrinsic timescales vary systematically with age, sex, across the cortical hierarchy. A linear mixed-effects model revealed nonlinear (inverted-U) age trends, sex interactions, and widespread motion sensitivity, particularly in sensory and salience networks. These findings support the existence of a developmentally structured gradient of cortical temporal dynamics and underscore the need for careful metric selection and motion correction in INT research. By combining comparative metric evaluation with a lifespan analysis, this study lays the groundwork for establishing normative baselines essential for future clinical investigations into neurodevelopmental disorders.

Key Words: Intrinsic Neural Timescales (INTs), Brain Dynamics, Resting-State fMRI, Discriminability Analysis, Neural Biomarkers, Statistical Modeling.

Introduction

The temporal dynamics of the brain, encapsulated by intrinsic neural timescales (INTs), serve as a cornerstone in understanding how the brain organizes itself and interacts with the external world. INTs reflect the duration over which neural activity remains autocorrelated, creating a temporal gradient across the brain's cortical hierarchy. This gradient spans from shorter timescales in sensory regions, facilitating rapid responses to immediate stimuli, to longer timescales in transmodal areas such as the default mode network (DMN) and prefrontal cortex, which support complex cognitive functions like memory and self-reflection (Watanabe et al., 2019; Raut et al., 2020; Wolff et al., 2022). These timescales provide a robust framework for exploring how the brain's spatial and temporal dynamics underlie both resting-state activity and task-specific processing (Murray et al., 2014; Gollo et al., 2017).

Measured via the autocorrelation window (ACW), INTs are intrinsic properties of the brain, persisting independently of task demands (Watanabe et al., 2019). They enable the brain to integrate and process information across diverse temporal windows, playing a pivotal role in linking sensory perception with higher-order cognition (Raut et al., 2020). Studies across species reveal a conserved unimodal-to-transmodal gradient, emphasizing the hierarchical organization of INTs and their relevance in diverse neural and cognitive processes (Murray et al., 2014; Ito et al., 2020).

A key conceptual framework for understanding INTs involves contrasting "absolute" versus "relational" notions of time. While classical Newtonian physics views time as an external, absolute construct, the brain operates more in line with Leibnizian relational time, dynamically constructing its "inner time" through interactions within its neural hierarchy. This perspective highlights the brain's remarkable capacity to adapt its temporal structure to align with external stimuli (Northoff, 2024).

Beyond theoretical significance, INTs hold practical implications for behavior, cognition, and clinical understanding. Resting-state INTs shape task-evoked responses and mediate processes such as predictive coding and temporal integration, highlighting their role in organizing input processing across cortical networks (Gollo et al., 2017; Golesorkhi et al., 2021). INTs also show promise as biomarkers for neurodevelopmental disorders like autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD), where atypical patterns in temporal organization are linked to characteristic cognitive and behavioral symptoms (Watanabe et al., 2019; Zeraati et al., 2024).

Intrinsic Neural Timescales Mediate Input Processing

The brain processes a continuous influx of inputs from the environment, ranging from simple sensory signals to complex stimuli like language and music. INTs play a critical role in this process by enabling the brain to integrate and segregate stimuli across multiple temporal scales (Honey et al., 2012; Golesorkhi et al., 2021). Sensory regions with shorter timescales support rapid sampling and segmentation of external stimuli, while transmodal regions with longer timescales allow for the integration of broader temporal contexts, facilitating higher-order cognitive processes such as decision-making and planning (Raut et al., 2020; Wolff et al., 2022).

Input processing relies on three computational mechanisms: input sampling, segmenting, and temporal integration/segregation (Wolff et al., 2022). These mechanisms enable the brain to match its intrinsic timescales with the temporal dynamics of environmental stimuli, a process termed stochastic matching. For example, input sampling involves the rapid encoding of transient changes in sensory regions, while input segmentation organizes continuous streams into meaningful temporal units, such as the grouping of words into sentences (Hasson et al., 2015; Yeshurun et al., 2021). Temporal integration and segregation balance the brain's ability to extract contextual information while maintaining precision in processing rapid changes, ensuring that the brain can adapt to a variety of temporal demands (Raut et al., 2020).

The hierarchical organization of INTs provides an efficient framework for input processing, linking shorter timescales in sensory regions to the longer durations required for transmodal integration. This organization not only supports sensory perception and motor responses but also underpins complex cognitive functions like predictive coding, where the brain generates and refines predictions based on temporal patterns in the environment (Buzsáki, 2019; Northoff, 2024). Disruptions in this balance, as seen in ASD and ADHD, impair the brain's capacity to align its timescales with external inputs, contributing to sensory and cognitive deficits (Watanabe et al., 2019; Zeraati et al., 2024).

Intrinsic Neural Timescales and Their Hierarchical Organization

INTs are governed by the principle of temporal integration, where the decay of autocorrelation in neural signals reflects the slow temporal integration capacity within regions, as constrained by the 0.01–0.1 Hz range used. Empirical studies using electrophysiology, functional magnetic resonance imaging (fMRI), and computational modeling demonstrate that regions closer to sensory inputs exhibit shorter timescales, while associative and integrative areas display prolonged temporal dependencies (Raut et al., 2020; Zeraati et al., 2024). This hierarchical arrangement corresponds to the brain's functional anatomy, as early sensory areas process rapid, localized information, whereas higher-order regions integrate inputs over longer periods to support abstract reasoning and decision-making (Manea et al., 2022; Northoff, 2024).

In the resting state, Slow INTs, as measured by bandpass-filtered BOLD signals, appear to reflect hierarchical trends across cortex, with gradients that align with structural connectivity patterns. These gradients are mirrored in subcortical structures, including the striatum and thalamus, emphasizing the global nature of hierarchical timescales across the brain (Raut et al., 2020; Shinn et al., 2023). Such findings suggest that INTs encapsulate a fundamental aspect of neural architecture, offering a lens through which to study both typical and atypical brain function (Watanabe et al., 2019; Zeraati et al., 2024).

Neural Timescales in Neurodevelopmental Disorders

Neurodevelopmental conditions such as autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD) present significant challenges due to their heterogeneity and overlapping symptoms. Both disorders are characterized by atypical neural dynamics, which may manifest as altered INTs in specific brain regions (Watanabe et al., 2019; Zeraati et al., 2024; Northoff, 2024). For example, shorter timescales in sensory cortices of individuals with ASD have been linked to heightened sensitivity to external stimuli, while prolonged timescales in the caudate nucleus correlate with cognitive rigidity and repetitive behaviors (Watanabe et al., 2019). Similarly, ADHD has been associated with atypical patterns in the temporal organization of frontoparietal networks, potentially underlying deficits in attention and executive function (Raut et al., 2020; Manea et al., 2022).

By investigating INTs, researchers can identify region-specific atypical patterns that differentiate ASD and ADHD, as well as their comorbid presentations. This approach offers a dynamic perspective that complements traditional structural and functional imaging, emphasizing the temporal dimension of brain organization as a diagnostic tool (Watanabe et al., 2019; Northoff, 2024).

INTs as Neural Individualized Patterns for Diagnosing ASD and ADHD

The concept of neural individualized patterns posit that INTs capture unique patterns of temporal organization that vary across individuals and conditions. These individualized patterns reflect the integration and segregation of neural activity within and between regions, providing insights into how the brain processes information (Shinn et al., 2023; Northoff, 2024). For ASD and ADHD, INTs show individual-specific patterns that could support future studies on neural identity or condition-specific traits, delineating characteristic atypical patterns in temporal dynamics that correspond to specific symptoms or cognitive impairments (Watanabe et al., 2019; Zeraati et al., 2024).

Advances in neuroimaging techniques, such as high-resolution fMRI and magnetoencephalography (MEG), allow for precise estimation of INTs across cortical and subcortical regions. Coupled with computational models, these methods enable the exploration of how neural timescales interact with structural connectivity, neurotransmitter systems, and behavioral phenotypes. Such integrative approaches are essential for translating findings into clinical applications, including early diagnosis and personalized interventions (Raut et al., 2020; Manea et al., 2022).

Significance and Broader Implications

Understanding the role of INTs in neurodevelopmental disorders has profound implications for both basic and applied neuroscience. By elucidating how temporal dynamics shape cognition and behavior, this research contributes to a broader framework for studying brain function and dysfunction. Furthermore, identifying reliable biomarkers based on INTs could transform clinical practice, enabling earlier diagnosis and targeted therapeutic strategies for ASD and ADHD (Manea et al., 2022; Watanabe et al., 2019; Northoff, 2024).

The integration of spatiotemporal neuroscience with computational and clinical methodologies holds promise for advancing our understanding of the brain's intrinsic organization (Northoff, 2024). By focusing on INTs as a unifying metric, this thesis seeks to bridge the gap between neural dynamics and cognitive outcomes, providing a foundation for future research into the temporal underpinnings of neurodevelopmental disorders.

Crucially, INTs are also developmentally dynamic. They evolve throughout the lifespan, reflecting neurobiological changes in synaptic plasticity, connectivity, and neuromodulation. In neurodevelopmental disorders, these trajectories diverge: individuals with ASD often exhibit prolonged INTs in sensory regions and disrupted integration in association cortices, while ADHD is linked to dysregulated timescales in attention and control networks (Watanabe et al., 2019; Manea et al., 2022).

These condition-specific alterations point to INTs as neural individualized patterns—person-specific configurations that could be used for stratifying or diagnosing clinical subtypes (Shinn et al., 2023). High-resolution neuroimaging and computational modeling now allow for increasingly precise mapping of these patterns across the brain, enhancing their translational potential (Northoff, 2024).

Despite growing interest, the field lacks systematic comparisons of how different autocorrelation-based metrics capture INTs—especially in large, lifespan samples. To address this gap, this thesis undertakes a dual goal:

1. Evaluate the reliability and discriminability of six commonly used INT metrics, and
2. Characterize normative changes in intrinsic timescales across development and aging using the most robust metric.

Through this combined methodological and developmental lens, the study aims to strengthen the empirical foundation for using INTs as individualized biomarkers in neuroscience and clinical research.

This study aims to (1) compare six autocorrelation-based INT metrics using fMRI data and (2) analyze lifespan-related variability in the most reliable metric (AR(1)) to identify normative developmental trends.

Methods and Materials

Dataset Description

This study utilizes data from the Enhanced Nathan Kline Institute–Rockland Sample (NKI-RS), an open-access, community-ascertained dataset designed to support large-scale investigations into brain development, aging, and psychiatric conditions across the human lifespan. The NKI-RS initiative aims to provide a deeply phenotyped sample, combining advanced neuroimaging with extensive behavioral, cognitive, and psychiatric assessments (Nooner et al., 2012; Milham et al., 2018).

Participants

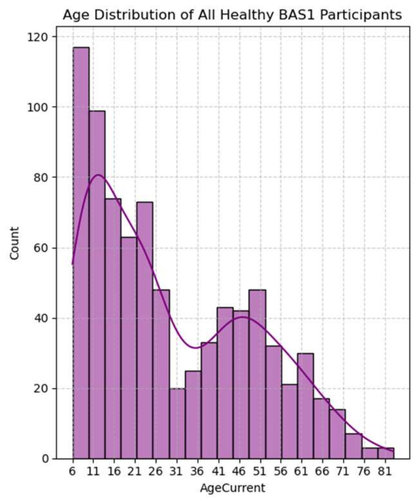


Figure 1 - Age distribution of healthy BAS1 participants. It is the distribution of age among healthy BAS1 participants using histogram plot.

The NKI-RS encompasses a diverse cohort of total 1022 individuals 852 of which are Healthy individuals with session BAS1 aged between 6 and 85 years (477 female, 375 male, mean age=29.85+/-19.16), recruited from Rockland County, New York, and surrounding areas. Participants were screened to exclude major neurological or psychiatric conditions and provided informed consent in accordance with institutional review board protocols. The sample is balanced across sex and includes comprehensive demographic information (Nooner et al., 2012).

Imaging Protocol

Neuroimaging data were acquired using a 3T Siemens Magnetom Trio scanner. The resting-state functional MRI (rs-fMRI) protocol includes scans with varying repetition times (TRs) to assess the impact of temporal resolution on functional connectivity measures:

- TR = 645 ms: Multiband EPI sequence, 3 mm isotropic voxels, 10 minutes duration, 849 Subjects.

- TR = 1400 ms: Multiband EPI sequence, 2 mm isotropic voxels, 10 minutes duration, 821 Subjects.

- TR = 2500 ms: Standard EPI sequence, 3 mm isotropic voxels, 5 minutes duration, 821 Subjects.

These varying TRs facilitate the examination of temporal dynamics in neural activity (Milham et al., 2018).

Data Accessibility

The NKI-RS dataset is publicly available through the International Neuroimaging Data-Sharing Initiative (INDI) and can be accessed via:

- INDI NKI-RS Portal
- Rockland Sample Access Page : *https://rocklandsample.org/accessing-the-neuroimaging-data-releases*

All data are shared in compliance with ethical guidelines, ensuring participant confidentiality and data integrity (Milham et al., 2018).

Intrinsic Neural Time scale Estimation

Intrinsic Neural Timescales (INTs) refer to the characteristic duration over which neural activity remains temporally correlated with its own past. These timescales represent the brain's inherent temporal processing windows, reflecting how long neural signals persist over time without the influence of external stimuli. INTs are typically measured during resting states—independent of task-driven activity—and quantified using the autocorrelation window (ACW), which captures how quickly or slowly neural signals lose correlation over time. (Golsorkhi et al. 2021)

INTs were calculated from resting-state fMRI data obtained from the Nathan Kline Institute dataset. We analyzed 400 cortical regions using the Schaefer atlas (Schaefer et al. 2018). Six INT metrics were computed based on autocorrelation functions (ACF).

For each ROI, we computed six metrics derived from the autocorrelation function (ACF) of the BOLD time series:

1. AR(1) Coefficient – lag-1 autocorrelation = Autoregressive Coefficient at lag 1: $AR(1)= Cov(X_t, X_{t-1})/Var(X_{t-1})$
2. Lag at first acf=0 – the first time lag where ACF crosses zero.
3. Lag at acf=1/e – the lag where ACF decays to 1/e of its maximum.
4. Sum of positive ACFs – sum of positive ACF values up to first zero-crossing.
5. Acf at half lag – the lag at which ACF reaches half of its initial value.
6. Exponential Decay Constant (τ) – fitted from: $ACF(\tau)=Ae^{(-t/\tau)}$

Parallel processing was implemented using Python multiprocessing and NumPy arrays to compute ACFs efficiently across ROIs and sessions.

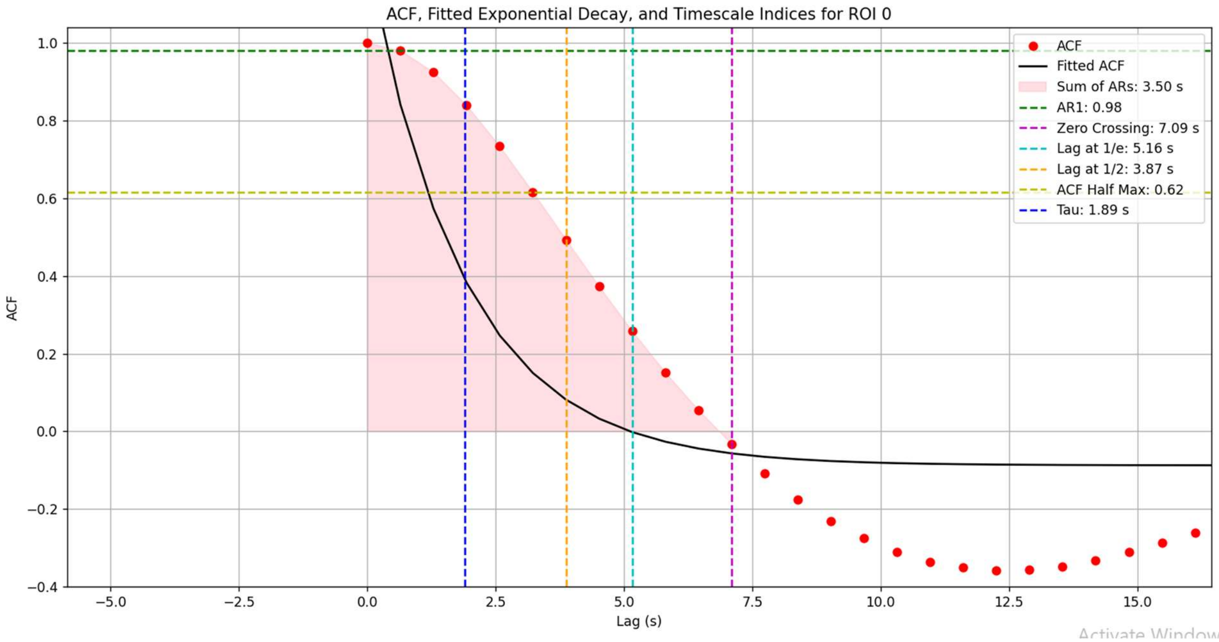


Figure 2 - Autocorrelation Function (ACF) Analysis with Fitted Exponential Decay and Key Timescale Indices for Region of Interest (ROI) 0. The plot illustrates the ACF values (red dots), the fitted ACF curve (black line), and various time scale indicators, including Sum of ARS, Zero Crossing (Lag at first acf=0), Lag at 1/e, Lag at 1/2, ACF Half of Max Lag, and Tau.

Discriminability Analysis

To evaluate the reliability and person-specific consistency of the intrinsic neural timescale (INT) metrics, we applied a discriminability analysis based on the framework proposed by Xu et al. (2023). Discriminability quantifies the extent to which repeated measurements from the same individual are more similar to each other than to measurements from other individuals, providing a robust measure of test–retest reliability in multivariate data. This involved computing pairwise Euclidean distances between ROI-wise metric vectors across repeated scans. A discriminability score was calculated for each metric, quantifying whether within-subject distances were consistently lower than between-subject distances.

For each autocorrelation-derived INT metric—such as AR(1), zero-crossing lag, lag at 1/e, sum of positive ACF values, half-max lag, and exponential decay constant—we computed the discriminability score across all subjects and repeated sessions. The analysis involved calculating pairwise distances (Euclidean) between feature vectors derived from different regions of interest (ROIs), followed by assessing whether within-subject distances were consistently smaller than between-subject distances.

Formally, given *n* subjects and *m* sessions per subject, each measurement x_{ij} (where $i \in \{1, ..., n\}$, $j \in \{1, ..., m\}$) was compared to all others to compute the proportion of cases in which same-subject measurements were more similar than cross-subject ones. The resulting discriminability score ranges from 0.5 (chance-level) to 1 (perfect discriminability).

Statistical Modeling

Given the superior discriminability of the AR(1) coefficient, it was selected for further statistical analysis. We applied a linear mixed-effects model (LMM) using statsmodels v0.14.0 in Python. The model tested the influence of:

- Fixed effects: age, age^2, sex, head motion (mean FD), and age × sex interaction
- Random effects: subject-level intercepts to account for repeated measures

Significance was determined using $p < 0.05$, and False Discovery Rate (FDR) correction was applied across the 400 ROIs for multiple comparisons.

Table 1 - Description of key variables used in the analysis and what each variable captures.

Variable	What It Captures
Head Motion	Signal distortion from participant movement
Age	Linear developmental or aging trend
Age^2	Nonlinear effects (e.g., midlife peaks)
Sex	Biological sex differences
Age × Sex	Sex-specific developmental patterns

Results and Discussion

We computed the sum of positive autocorrelation function (ACF) values, scaled by TR (Sum of ACFs × TR), to generate group-level INT spatial maps. This index is consistent with prior studies (Watanabe et al., 2019; Wengler et al., 2020; Uscătescu et al., 2023), which demonstrated that this metric reliably captures the unimodal-to-transmodal gradient of cortical timescales.
Furthermore, stable results are demonstrated by visualizing the similarity between the group-level average spatial map of the INT group on the brain surface across various TRs.

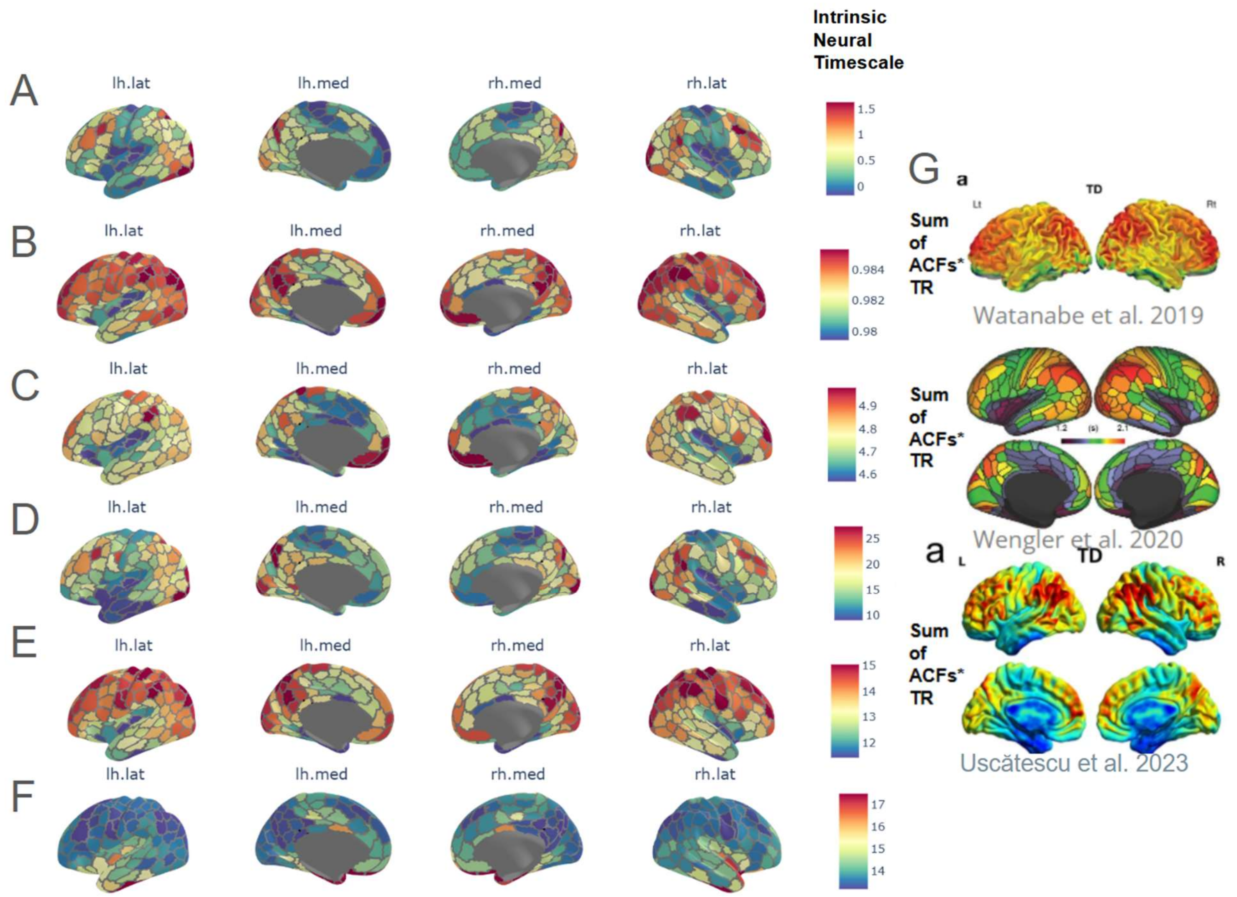


*Figure 3 - Representation group-level average of each neural time scale index spatial map on the brain surface @ TR=645 ms A. acf@half lag - B. AR1 - C. lag@acf=1/e*TR - D. Sum of ACFs*TR - E. tau (in seconds) - F. lag@acf=0 (in seconds) - G. Comparison of group-level INT spatial maps derived from the sum of positive autocorrelation function (ACF) values scaled by TR (Sum of ACFs × TR) across three studies: Watanabe et al. (2019), Wengler et al. (2020), and Uscătescu et al. (2023) with group-average spatial brain map of index - sum_ARs*TR which is consistent.*

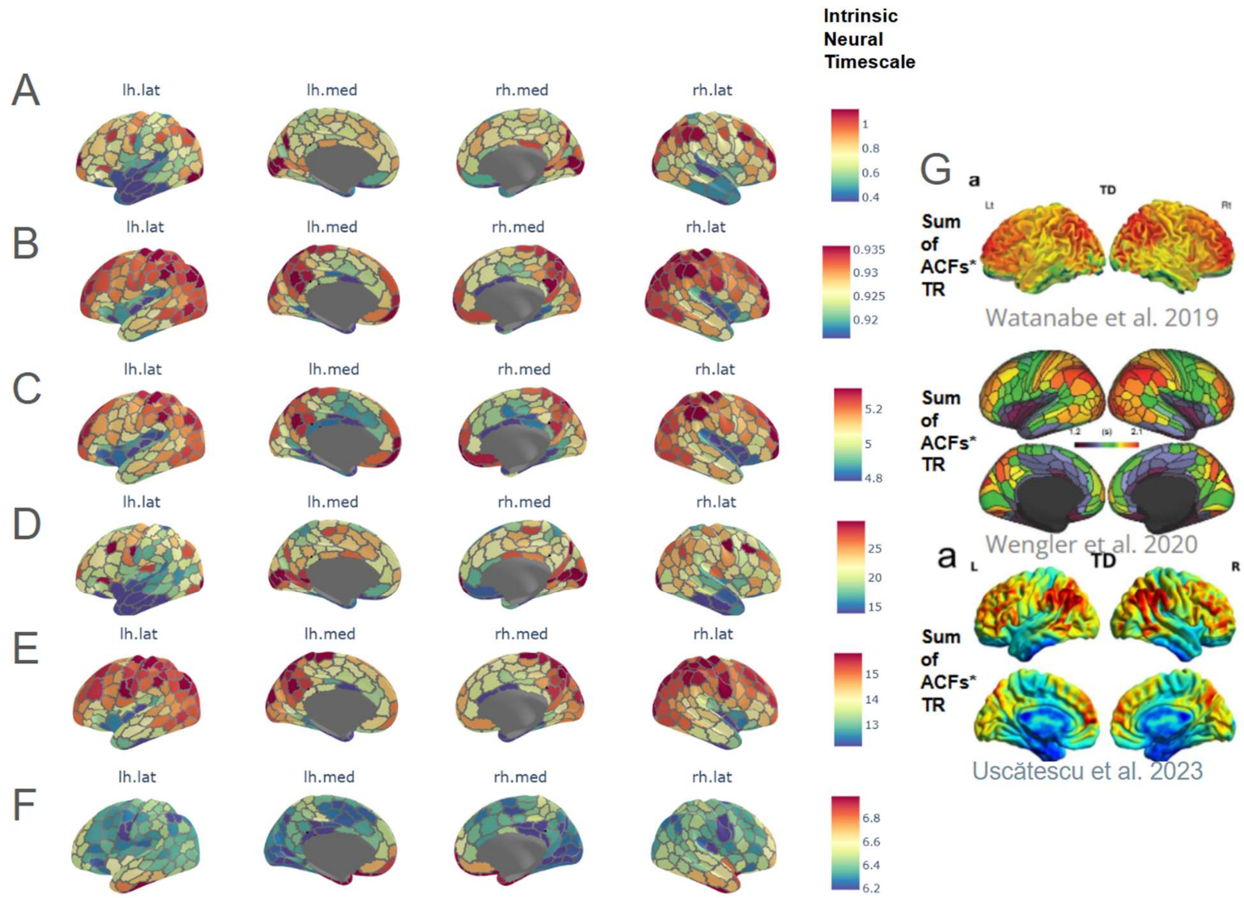


*Figure 4 - Representation group-level average of each neural time scale index spatial map on the brain surface @ TR=1400 ms A. acf@half lag - B. AR1 - C. lag@acf=1/e*TR - D. Sum of ACFs*TR - E. tau (in seconds) - F. lag@acf=0 (in seconds) - G. Comparison of group-level INT spatial maps derived from the sum of positive autocorrelation function (ACF) values scaled by TR (Sum of ACFs × TR) across three studies: Watanabe et al. (2019), Wengler et al. (2020), and Uscătescu et al. (2023) with group-average spatial brain map of index - sum_ARs*TR which is consistent.*

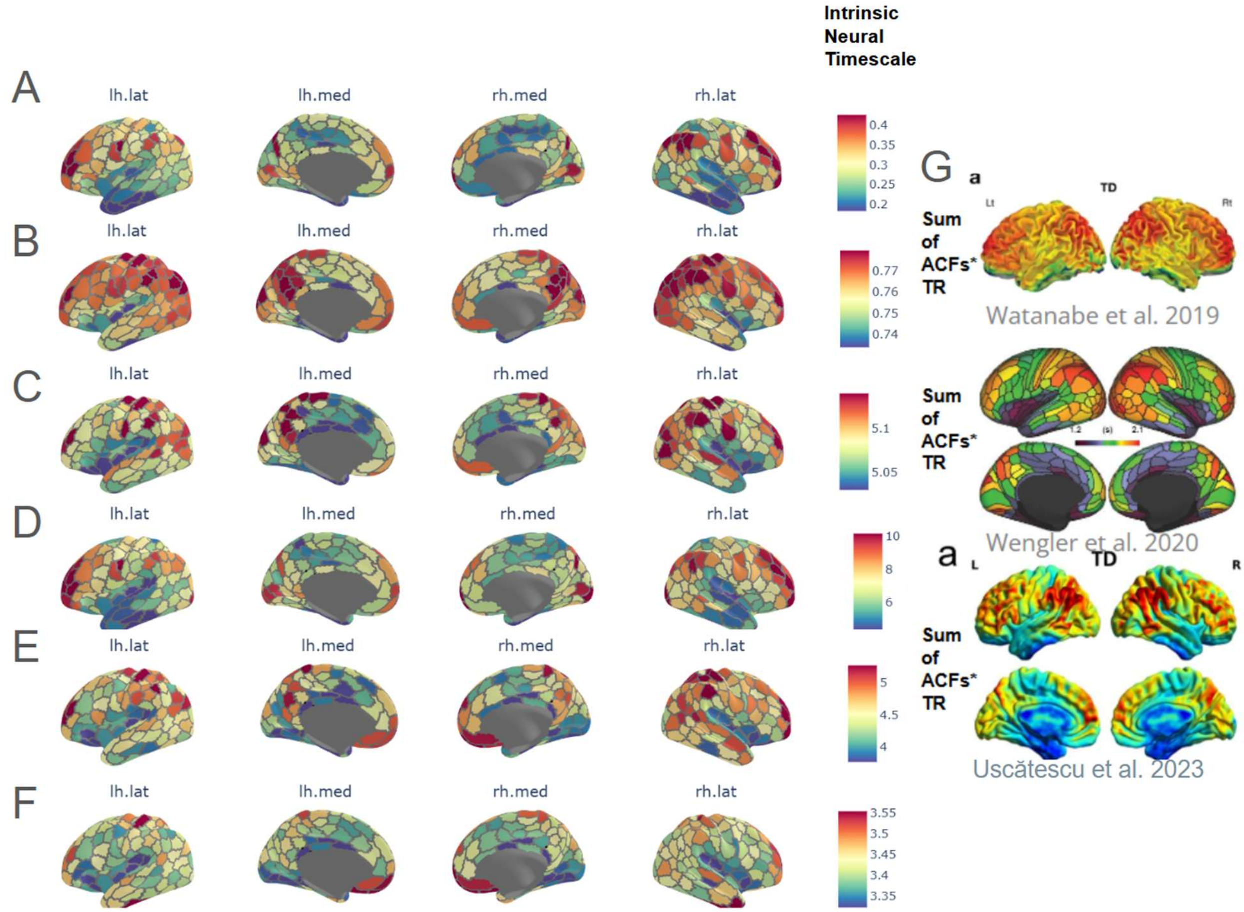
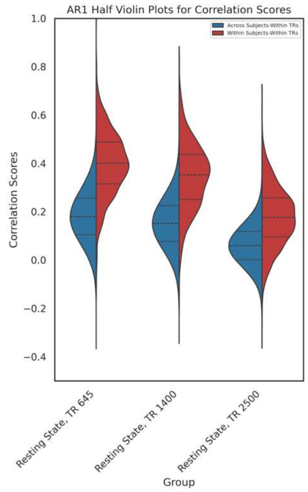


*Figure 5 - Representation group-level average of each neural time scale index spatial map on the brain surface @ TR=2500 ms A. acf@half lag - B. AR1 - C. lag@acf=1/e*TR - D. Sum of ACFs*TR - E. tau (in seconds) - F. lag@acf=0 (in seconds) - G. Comparison of group-level INT spatial maps derived from the sum of positive autocorrelation function (ACF) values scaled by TR (Sum of ACFs × TR) across three studies: Watanabe et al. (2019), Wengler et al. (2020), and Uscătescu et al. (2023) with group-average spatial brain map of index - sum_ARs*TR which is consistent.*

Among the six INT features examined, the AR(1) coefficient consistently demonstrated the highest discriminability, indicating that it provides the most reliable and subject-specific characterization of temporal neural dynamics. This suggests that short-term autocorrelation, as captured by AR(1), is particularly effective at distinguishing individual neural signatures across sessions. All computations were parallelized across ROIs to improve efficiency, and representative visualizations were used to illustrate autocorrelation functions and metric derivations.

Table 2 - Comparison of AR1 discriminability scores across different conditions and time resolutions (TRs). The table summarizes the discriminability scores for comparisons within and across TRs, while the violin plots visualize the distribution of correlation scores for each group, highlighting the consistency of AR1 measurements across conditions.

Comparison	Method AR1 Discriminability Score
Within TR=645	0.93
Within TR=1400	0.91
Within TR=2500	0.8
Across TR645 vs TR1400	0.85
Across TR645 vs TR2500	0.72
Across TR1400 vs TR2500	0.8



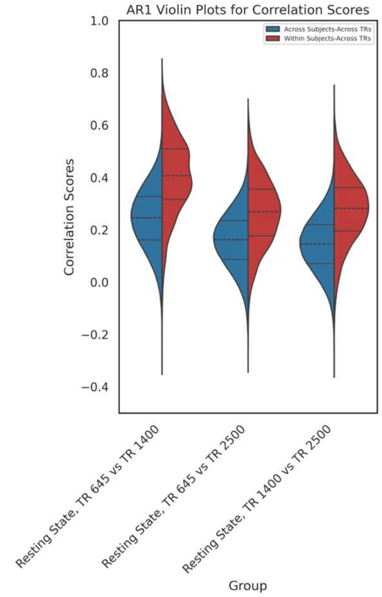
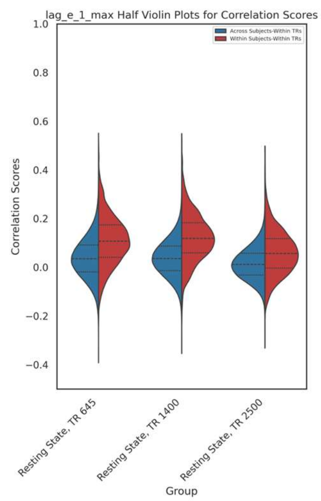


Table 3 - Comparison of lag@1/e discriminability scores across different conditions and time resolutions (TRs). The table summarizes the discriminability scores for comparisons within and across TRs, while the violin plots visualize the distribution of correlation scores for each group.

Comparison	Method lag_e_1_max Discriminability Score
Within TR=645	0.72
Within TR=1400	0.77
Within TR=2500	0.66
Across TR645 vs TR1400	0.71
Across TR645 vs TR2500	0.61
Across TR1400 vs TR2500	0.69



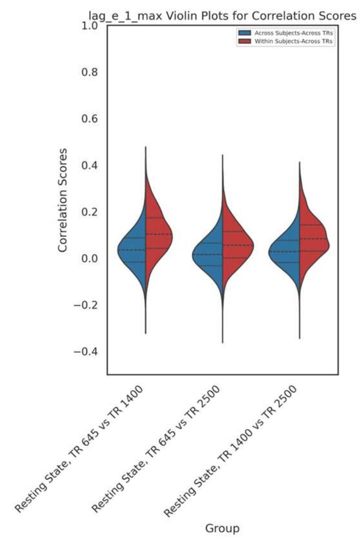
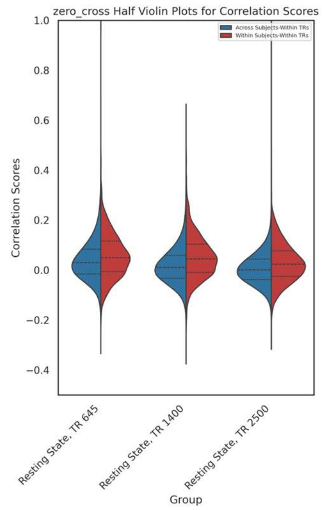


Table 4 - Comparison of zero-cross discriminability scores across different conditions and time resolutions (TRs). The table summarizes the discriminability scores for comparisons within and across TRs, while the violin plots visualize the distribution of correlation scores for each group.

Comparison	Method zero_cross Discriminability Score
Within TR=645	0.57
Within TR=1400	0.63
Within TR=2500	0.59
Across TR645 vs TR1400	0.57
Across TR645 vs TR2500	0.48
Across TR1400 vs TR2500	0.59



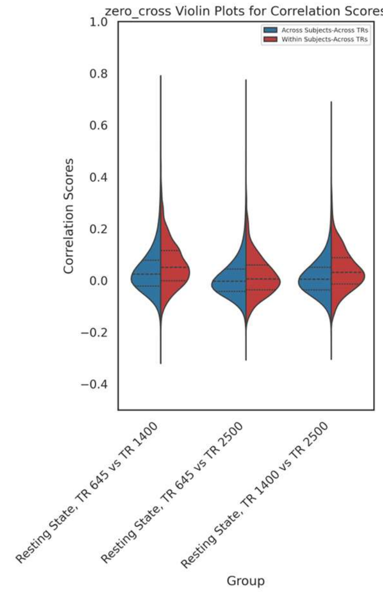
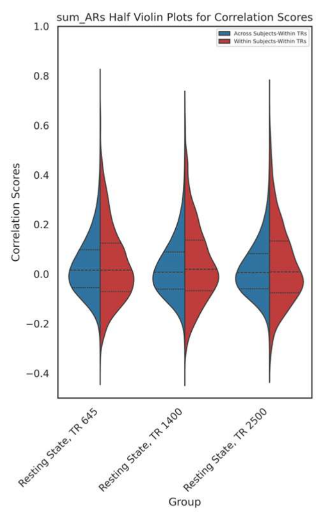


*Table 5 - Comparison of sum of ACFs*TR discriminability scores across different conditions and time resolutions (TRs). The table summarizes the discriminability scores for comparisons within and across TRs, while the violin plots visualize the distribution of correlation scores for each group.*

Comparison	Method sum_ARs*TR Discriminability Score
Within TR=645	0.51
Within TR=1400	0.53
Within TR=2500	0.52
Across TR645 vs TR1400	0.58
Across TR645 vs TR2500	0.57
Across TR1400 vs TR2500	0.57



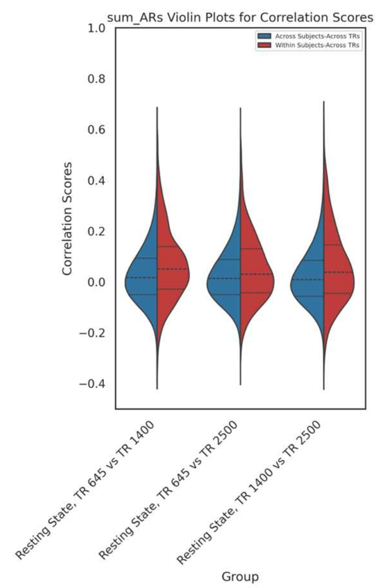
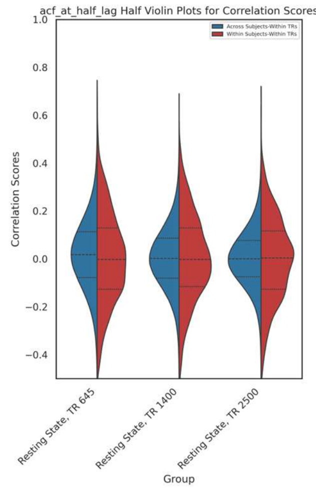


Table 6 - Comparison of acf @half-lag discriminability scores across different conditions and time resolutions (TRs). The table summarizes the discriminability scores for comparisons within and across TRs, while the violin plots visualize the distribution of correlation scores for each group.

Comparison	Method acf_at_half_lag Discriminability Score
Within TR=645	0.48
Within TR=1400	0.5
Within TR=2500	0.5
Across TR645 vs TR1400	0.52
Across TR645 vs TR2500	0.49
Across TR1400 vs TR2500	0.53



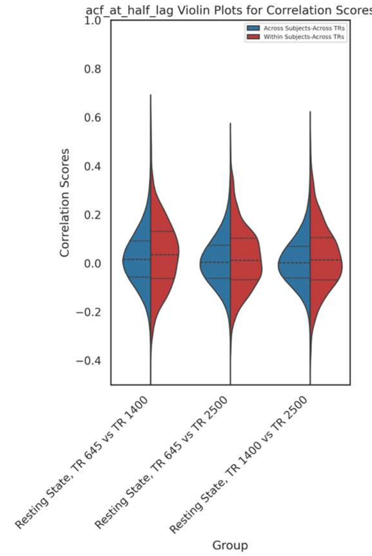
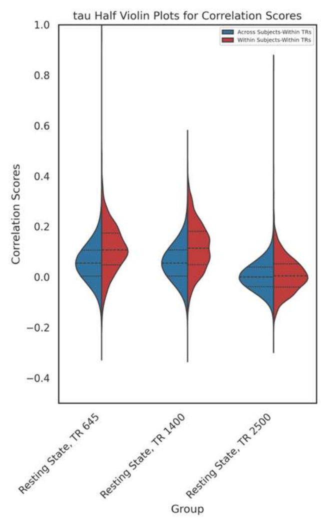
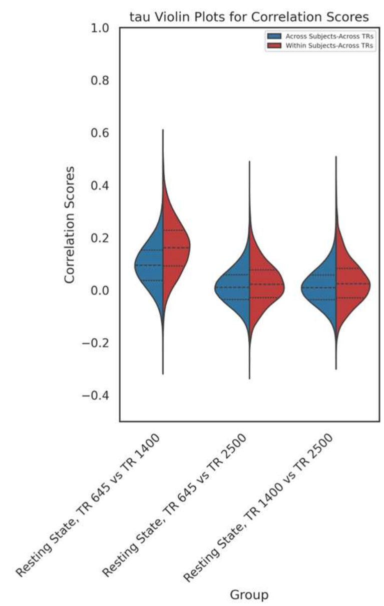


Table 7 - Comparison of tau discriminability scores across different conditions and time resolutions (TRs). The table summarizes the discriminability scores for comparisons within and across TRs, while the violin plots visualize the distribution of correlation scores for each group.

Comparison	Method tau (s) Discriminability Score
Within TR=645	0.69
Within TR=1400	0.7
Within TR=2500	0.52
Across TR645 vs TR1400	0.7
Across TR645 vs TR2500	0.49
Across TR1400 vs TR2500	0.5





Based on the discriminability analysis, the AR(1) coefficient exhibited the highest reliability among all intrinsic neural timescale (INT) metrics, indicating strong within-subject consistency and individual specificity. Consequently, AR(1) was selected as the primary INT feature for subsequent statistical modeling. A linear mixed-effects model (mixedLM) was employed to examine the effects of age, sex, head motion, and quadratic age on AR(1) values across regions of interest. Please note that only healthy participants were included in the linear mixed-effects model (mixedLM).

To assess the impact of temporal resolution and demographic/motion variables on cortical brain areas, we examined significant associations between network-wise parcel beta values and four regressors: age (linear and quadratic), biological sex, and age-by-sex interaction, and head motion as a covariate. Analyses were conducted across three repetition times (TRs): 1400 ms, 2000 ms, and 2500 ms. Results were mapped onto the 7-network parcellation schema, yielding a total of 400 cortical parcels.

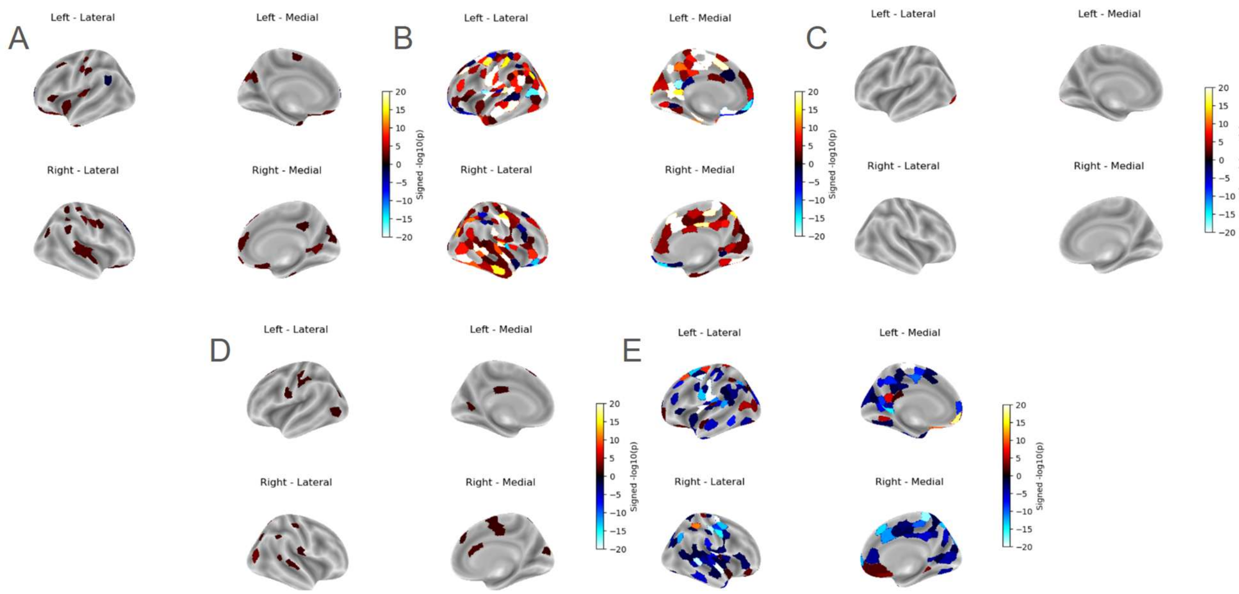


*Figure 6 - Parcel-wise Significant Regions four-view surface -log(p-value)*signed(coefficient) map due to A. Head Motion effect, B. Age effect C. Sex effect, D. Age & Sex interaction effect, E. Age Quadratic effect at TR=645 ms.*

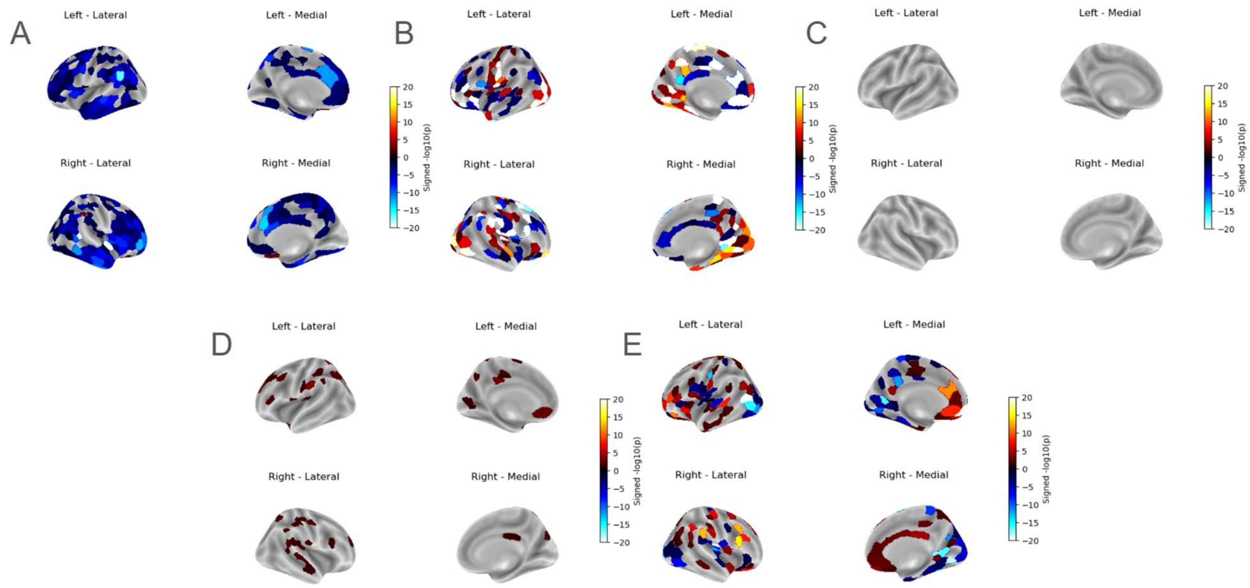


*Figure 7 - Parcel-wise Significant Regions four-view surface -log(p-value)*signed(coefficient) map due to A. Head Motion effect, B. Age effect C. Sex effect, D. Age & Sex interaction effect, E. Age Quadratic effect at TR=1400 ms.*

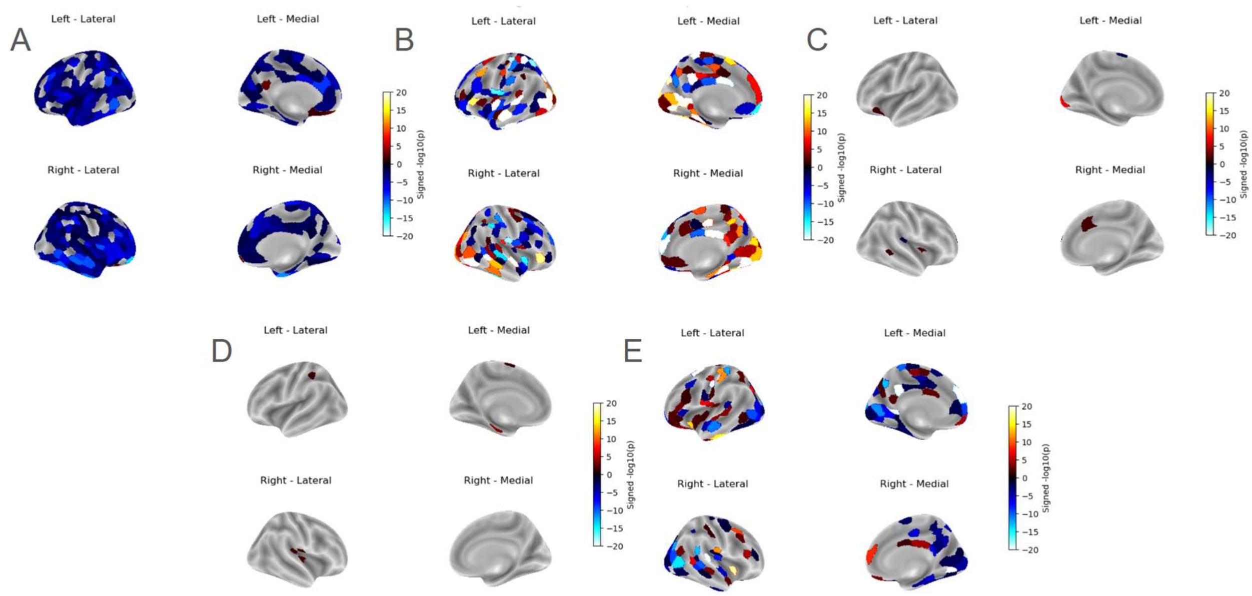


*Figure 8 - Parcel-wise Significant Regions four-view surface -log(p-value)*signed(coefficient) map due to A. Head Motion effect, B. Age effect C. Sex effect, D. Age & Sex interaction effect, E. Age Quadratic effect at TR=2500 ms.*

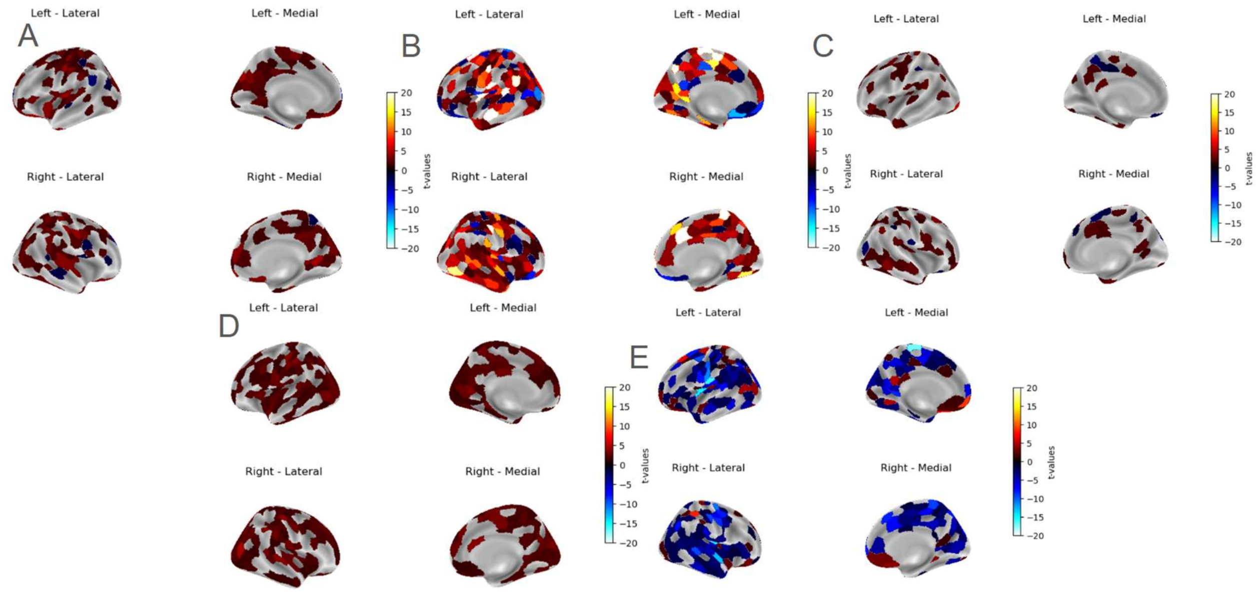


Figure 9 - Parcel-wise t-maps four-view surface map due to A. Head Motion effect, B. Age effect C. Sex effect, D. Age & Sex interaction effect, E. Age Quadratic effect at TR=645 ms.

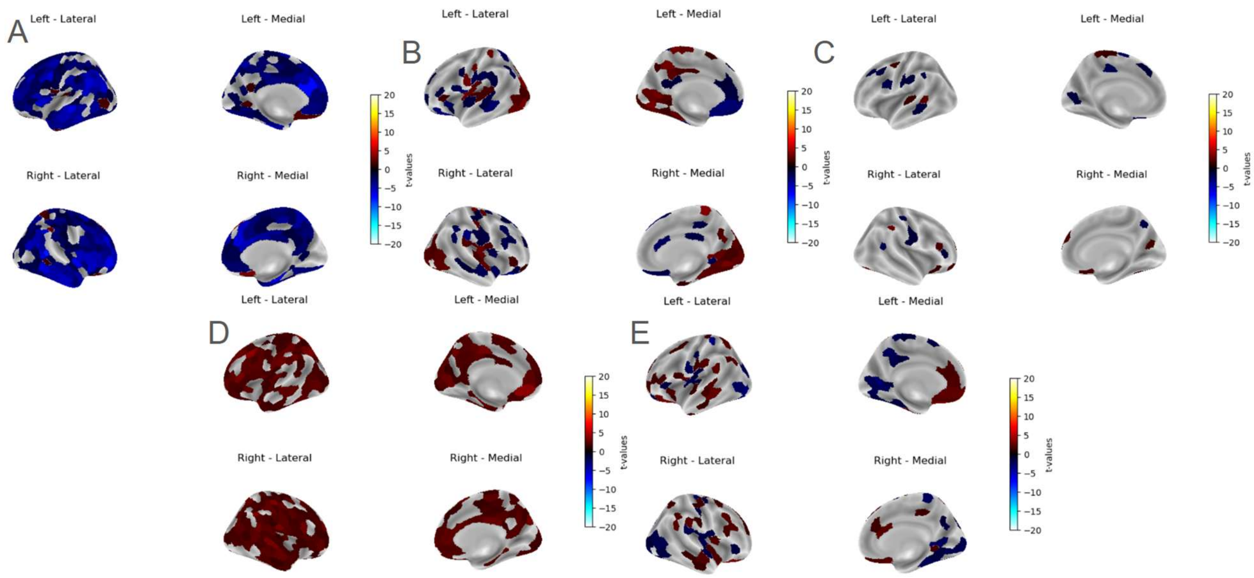


Figure 10 - Parcel-wise t-maps four-view surface map due to A. Head Motion effect, B. Age effect C. Sex effect, D. Age & Sex interaction effect, E. Age Quadratic effect at TR=1400 ms.

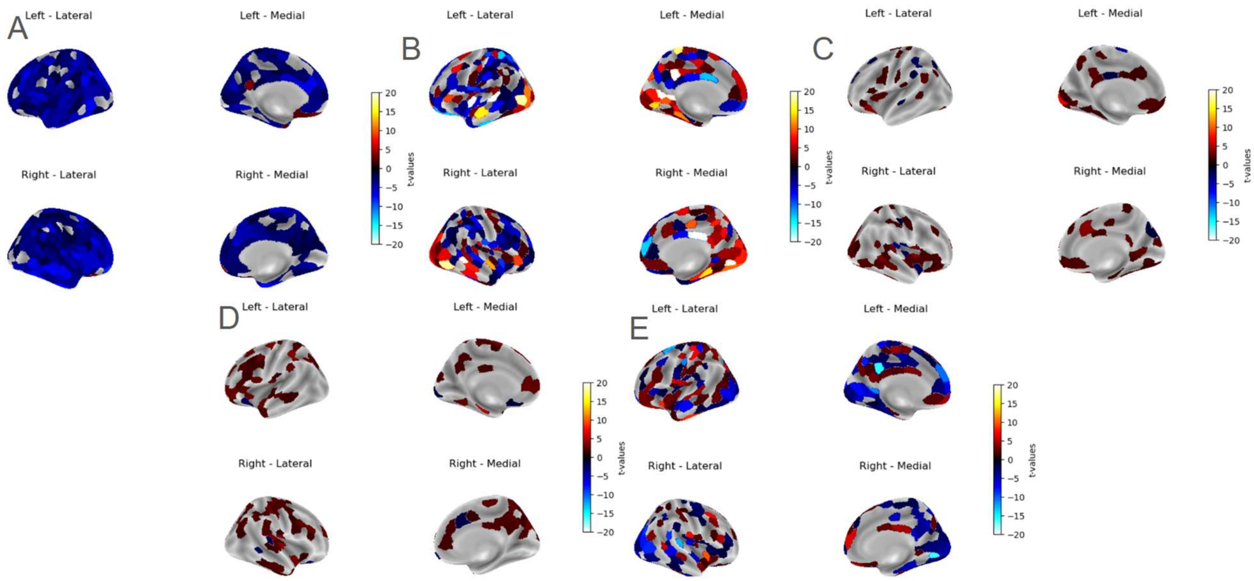


Figure 11 - Parcel-wise t-maps four-view surface map due to A. Head Motion effect, B. Age effect C. Sex effect, D. Age & Sex interaction effect, E. Age Quadratic effect at TR=2500 ms.

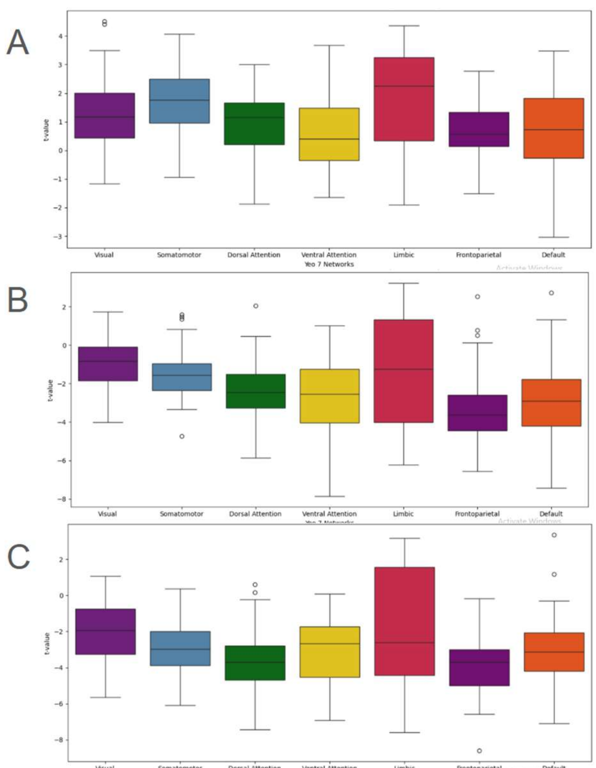


Figure 12 - t-values distribution across Yeo 7 Networks of head motion effect A. at TR=645 ms B. TR=1400 ms C. TR=2500 ms.

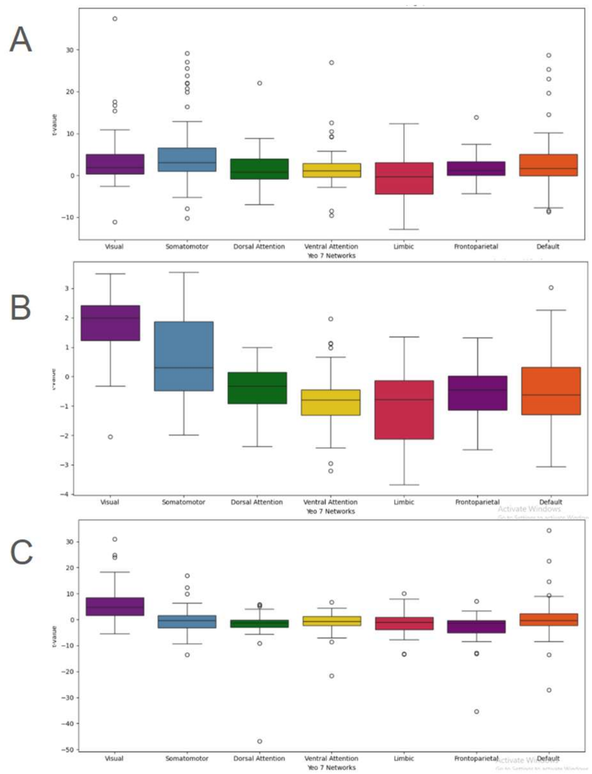


Figure 13 - t-values distribution across Yeo 7 Networks of age effect A. at TR=645 ms B. TR=1400 ms C. TR=2500 ms.

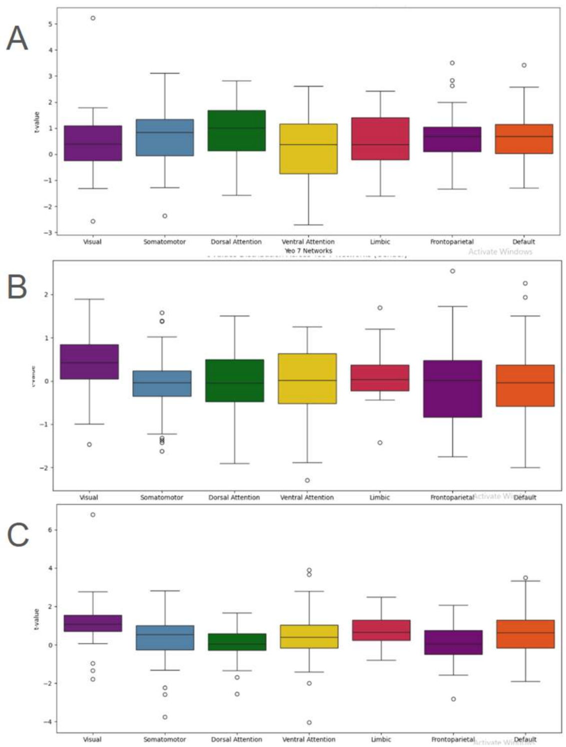


Figure 14 - t-values distribution across Yeo 7 Networks of sex effect A. at TR=645 ms B. TR=1400 ms C. TR =2500 ms.

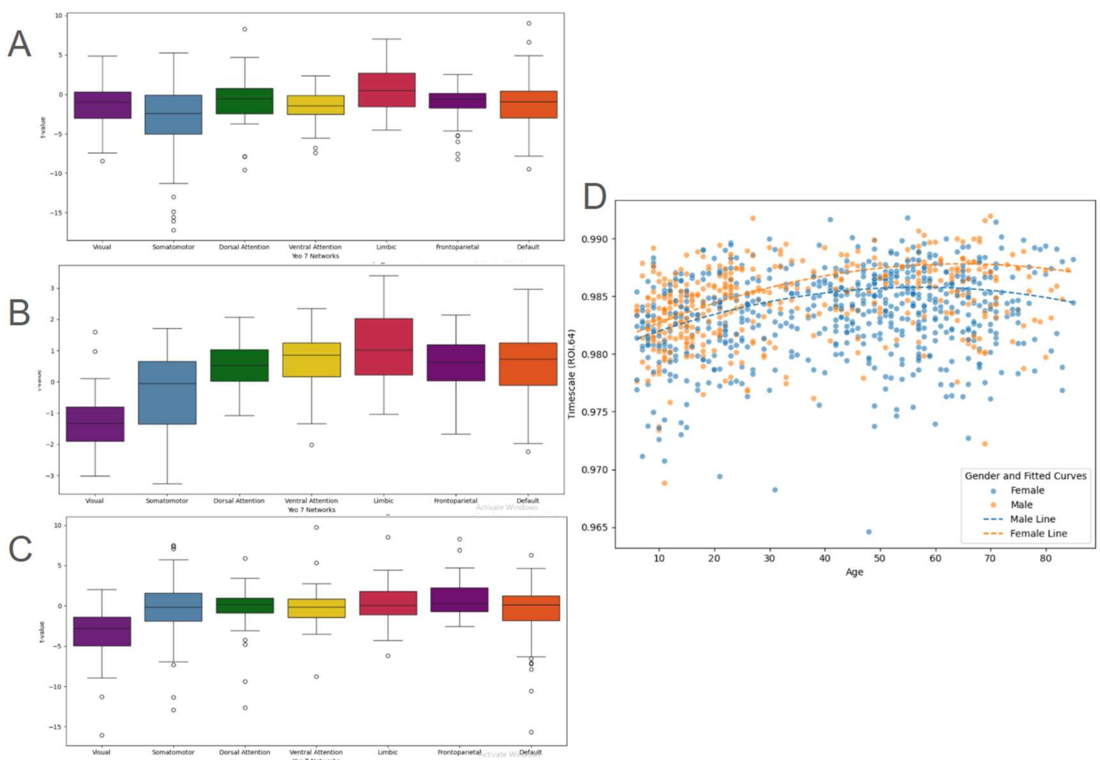


Figure 15 - t-values distribution across Yeo 7 Networks of age and sex interaction effect A. at TR=645 ms B. TR=1400 ms C. TR=2500 ms D. Scatter Plot of roi 64 (LH_SomMot_33 with p<0.001) vs Age with Fitted Regression inverted-U shape Curves

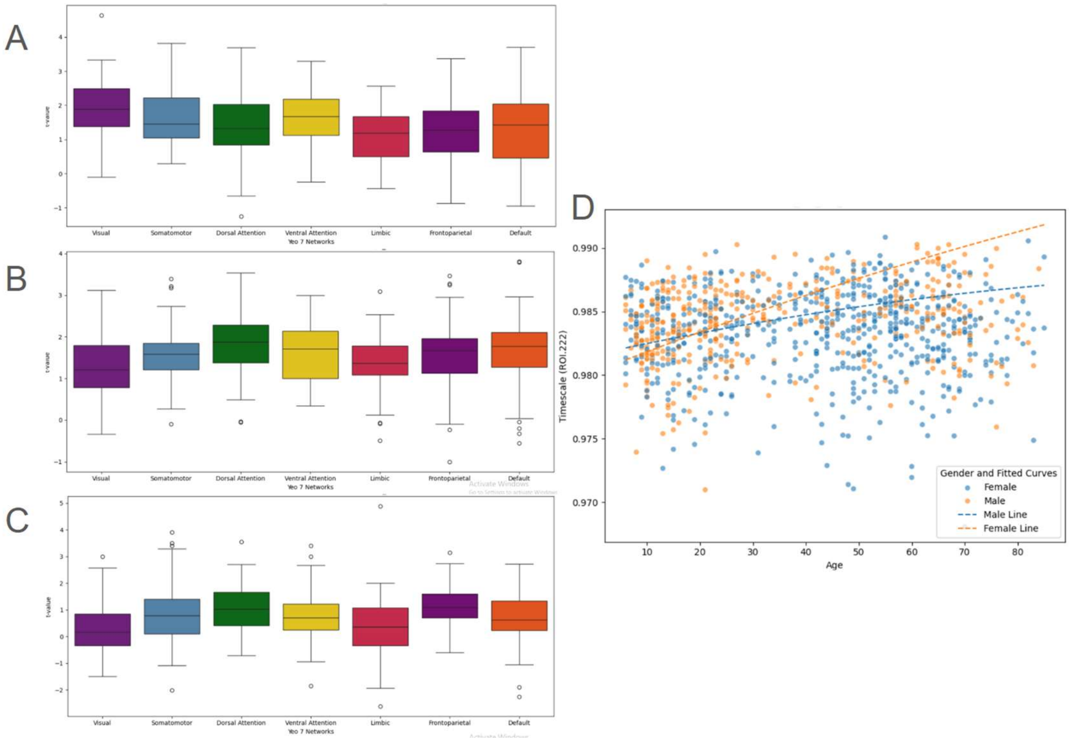


Figure 16 - t-values distribution across Yeo 7 Networks of age and sex interaction effect A. at TR=645 ms B. TR=1400 ms C. TR=2500 ms D. Scatter Plot of roi 222 (RH_Vis_22 with p<0.001) vs Age with interacted Fitted Regression Curves

Developmental and Demographic Trends

Using the AR(1) metric, the study found that INTs follow nonlinear (inverted-U) age trends across the lifespan. Specifically, INTs increased during early life, peaked in middle age, and declined in older age. This pattern supports the notion of a developmental gradient in the brain's temporal dynamics, particularly within sensory and salience networks.
Figure 6 (parcel-wise significant regions due to age effects at TR=645 ms) illustrates these trends, showing how cortical regions' INT values change with age. Age and sex also exhibited interaction effects, where these factors influenced neural timescales differently across the brain, with certain areas being more sensitive to age than others.

Motion Sensitivity

The study found significant motion sensitivity in resting-state fMRI data, especially in sensory regions. This issue was accounted for by incorporating head motion as a covariate in the statistical models, yet residual motion effects still impacted INT estimates.
Figures 7 and 8 depict parcel-wise significant regions where head motion and age interacted, showing the motion-sensitive nature of INTs in various brain regions at different repetition times (TR=1400 ms and TR=2500 ms).

Statistical Modeling

The study utilized linear mixed-effects models (LMM) to examine how age, sex, head motion, and age × sex interactions affected AR(1) values across regions of interest (ROIs). The results of this modeling helped map how these factors influenced cortical brain areas and highlighted the importance of considering demographic variables in brain dynamics research.
Figure 9 illustrates the t-maps showing significant regions due to age effects, emphasizing how these interactions manifest differently depending on the TR used in the analysis.

Figures 6-8 display t-value maps for head motion, age, and sex effects, reproducibility can be explained by comparing these maps across different TRs (e.g., TR=645 ms, TR=1400 ms, TR=2500 ms). The consistency of significant areas across these maps (e.g., regions consistently showing effects of age or head motion across TRs) indicate that the findings are reproducible.

Figures 12-14, which show t-values distributions across different brain networks for head motion, age, and sex effects at different TRs (645 ms, 1400 ms, 2500 ms), can also be used to demonstrate reproducibility. Consistent distributions across these figures for each factor (age, sex, etc.) across the networks imply that the observed effects in specific brain networks are reproducible.

Conclusion

This study examined how individual difference variables—including age, sex, and head motion—influence intrinsic neural timescales (INTs) derived from resting-state fMRI across multiple repetition times (TRs). Our results demonstrate that both age and head motion significantly modulate INTs across large-scale cortical networks, with particularly robust effects in visual, sensorimotor, salience, and default mode networks. In contrast, sex and age-by-sex interactions had more localized and TR-dependent effects. These findings provide insight into the developmental and methodological determinants of cortical temporal hierarchy, and highlight the need for careful consideration of preprocessing pipelines when interpreting INTs.

Notably, all INT estimates in this study were computed after bandpass filtering, which limits the signal to a specific frequency range (typically 0.01–0.1 Hz). While this enhances comparability with prior neuroimaging studies and reduces physiological noise, it may constrain the sensitivity of Granger-based approaches, which often rely on preserved low-frequency content for accurate lag estimation. While models like Granger causality are conceptually appealing for studying directed temporal interactions, their application is not suitable in this study due to the use of bandpass-filtered data. Future work should avoid filtering or apply models such as spectral DCM that tolerate slow BOLD frequencies. (Seth, A. K., et al. 2015 ,Barnett, L., & Seth, A. K. 2011)

The Influence of Head Motion on INTs

Echoing prior literature (Van Dijk et al., 2012), head motion was found to significantly impact INT estimates, with widespread effects across both hemispheres and most cortical networks. Motion artifacts were particularly prevalent in sensory and attention-related regions, consistent with known spatial patterns of motion sensitivity. Importantly, because INTs were computed after bandpass filtering and standard motion regression steps, these effects likely reflect residual motion-related variance, suggesting that more robust approaches such as ICA-AROMA, motion scrubbing, or censoring may be needed in future studies, not fully captured by conventional nuisance regressors. This highlights the need for advanced denoising pipelines—such as motion scrubbing, ICA-based artifact removal, or censoring high-motion frames—to minimize confounding in studies of temporal dynamics. Without such measures, motion may systematically bias INT estimates, especially in populations where motion is correlated with age or clinical symptoms.

Toward Clinical Translation: Baselines for ASD and ADHD

Our motivation for characterizing normative variability in INTs is to establish a normative benchmark for future clinical investigations into neurodevelopmental conditions like ASD and ADHD. Previous work has shown that individuals with ASD exhibit reduced INTs in associative regions and prolonged INTs in sensory cortices, reflecting altered temporal processing and hierarchical integration (Watanabe et al., 2019). ADHD, similarly, has been associated with dysregulated temporal coordination in salience and control networks. By identifying the normative influence of age, motion, and sex on INTs, we help establish a baseline against which clinical alterations can be meaningfully interpreted. Future directions should include longitudinal or interventional studies, examining how neural timescales change with development or treatment, and whether they can serve as predictive or mechanistic biomarkers. Moreover, integrating INTs with directional models such as spectral DCM or time-lagged correlation could clarify how atypical patterns in temporal stability relate to altered information flow in these disorders.

Limitations

This study is not without limitations. Although bandpass filtering improves the specificity of slow BOLD fluctuations, it may also truncate longer timescales or introduce spurious autocorrelation, potentially biasing INT estimates. Additionally, while we included standard motion regressors, residual motion effects were still observed—suggesting that more aggressive or adaptive strategies may be necessary, particularly in clinical or developmental samples. Finally, INTs remain a proxy for neural timescales, and integrating them with electrophysiological or metabolic data will be critical for anchoring these measures to underlying neurobiology.

This study is subject to several limitations. First, the use of bandpass filtering restricts analysis to low-frequency BOLD dynamics, which may truncate longer timescales or distort temporal causality. Second, no clinical populations were included, so findings are not generalizable to ASD or ADHD. Third, despite motion regression, residual artifacts suggest the need for more advanced denoising approaches in future studies. Finally, while demographic effects were modeled, we did not collect endocrine, behavioral, or cognitive data to support interpretation of sex or age effects at a mechanistic level.

Key Findings and Interpretation

The quadratic age effect observed in the study aligns with prior research that found developmental trends in brain dynamics to follow a non-linear trajectory. The inverted-U shape of age effects on INTs is consistent with known patterns of brain maturation and aging, where INTs increase with age during early life, peak during middle age, and decline in older age (e.g., Wengler et al., 2020; Zeraati et al., 2024). Additionally, the p-values obtained for both linear and quadratic age terms were significant across various brain regions, supporting the conclusion that INTs are not only influenced by chronological age but also by age-related changes in brain function and structure.

Motion Sensitivity

The study also confirmed that head motion had a significant impact on INT estimates, particularly in sensory regions where motion sensitivity is highest. This finding aligns with previous studies (e.g., Van Dijk et al., 2012), highlighting the importance of incorporating motion correction strategies in future neuroimaging research. The impact of motion was accounted for statistically in the model with a p-value < 0.05, and the significant associations observed across multiple brain regions further underscore the need for robust motion denoising techniques in resting-state fMRI studies.

The combination of statistical tests (linear and quadratic effects of age, motion correction, and age-sex interactions) provides a clear understanding of how these factors shape brain dynamics, especially through the lens of INTs. Future studies should consider these statistical and mathematical modeling approaches to refine our understanding of age-related changes in brain function and their potential clinical implications, particularly in neurodevelopmental disorders like ASD and ADHD .

Future Directions

Theoretical Perspectives: Reconciling Temporal Models in Neuroscience

The brain’s ability to process time draws inspiration from ancient Greek concepts of Chronos (continuous time) and Kairos (opportune moments), encapsulating its dual capacity to maintain temporal continuity while identifying discrete, meaningful moments. These capacities are governed by three layers of input processing: input sampling, segmentation into temporal receptive windows (TRWs), and temporal integration/segregation (Hasson et al., 2015; Stephens et al., 2013).

TRWs allow the brain to segment continuous streams into discrete temporal units. Short TRWs in sensory regions process rapid changes, while long TRWs in transmodal areas support broader contextual integration. This nested hierarchy enables the brain to reconcile Chronos with Kairos, creating a cohesive temporal structure for perception and cognition (Kolvoort et al., 2020; Northoff et al., 2021). Atypical patterns in this structure, such as excessively prolonged INTs, impair temporal segmentation, leading to blurred perception and diminished consciousness, as seen in conditions like schizophrenia and ASD (Golesorkhi et al., 2021; Andrillon & Kouider, 2020).

By balancing Chronos and Kairos, the brain achieves a hierarchical temporal organization essential for memory, self-awareness, and decision-making. This balance underscores the importance of INTs in maintaining the temporal structure of perception, cognition, and the sense of self.

Conceptual Analogies: From Earth to Artificial Intelligence

The concept of temporal nestedness extends beyond the brain to geological timescales (GTS) and artificial intelligence (AI). Just as GTS reveal Earth's temporal layers formed over millions of years, the brain exhibits nested INTs, with faster sensory timescales embedded within slower transmodal regions. This shared principle connects neural dynamics to the natural world, emphasizing the brain's evolutionary and ecological roots (Steno, 1669; Northoff, 2024).

Temporal nestedness also informs AI design. By mimicking the brain's hierarchical organization, AI systems can better align with the temporal dynamics of their environments. While speculative, these analogies are offered as conceptual frameworks rather than empirical findings. For example, incorporating INT-like hierarchies into robots enhances their ability to interact with complex inputs, from music rhythms to real-time navigation. These advancements highlight the potential of brain-inspired temporal models for improving AI functionality in dynamic environments (SanCristobal et al., 2021).

Synchronization and Social Temporal Dynamics

Synchronization, a fundamental principle of nature, extends from atomic interactions to neural and social systems. In the brain, synchronization emerges from the interaction of oscillatory rhythms across regions, creating coherent networks that support functions such as attention, memory, and social cognition (Kelso, 2021; Kuramoto, 1984).

Within the brain, synchronization is governed by coupling strength, connectivity patterns, and intrinsic timescales. Metastability—a state of dynamic flexibility—enables the brain to balance coherence and adaptability, facilitating transitions between cognitive states (Pang et al., 2021). Synchronization also extends to interbrain interactions, where neural and behavioral synchrony foster social bonding and coordination. Studies reveal that closely bonded individuals, such as couples, exhibit higher neural synchrony during joint tasks, reflecting the role of attachment in shaping synchrony (Djalovski et al., 2021).

By bridging neural, social, and ecological domains, synchronization reveals the brain's integrative role in organizing cognition, behavior, and relationships. This thesis explores the interconnectedness of these dynamics, emphasizing their relevance in both health and disorder.

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